



TECHNISCHE
UNIVERSITÄT
WIEN

CERTIFICATION

The Vice Rector Academic Affairs
as the governing body responsible for study matters awarded, pursuant to
§ 87 (1) University Act 2002, BGBl. I Nr. 120/2002 in the applicable version, to

Mr

Vincent Theo Willem KENBEEK BSc

born on 11.07.1996,
citizenship Netherlands,
who on 26.11.2025 finished his studies for the

Master Programme Computer Engineering

pursuant to the curriculum, MBl.Nr. 142-2011 in the applicable version,
by passing the compulsory examinations and master thesis
the academic degree of

DIPLOM-INGENIEUR

(Dipl.-Ing. or DI)

Equivalent to Master of Science (MSc)

Conferred in Vienna, on 26.11.2025

By order of the Vice Rector Academic Affairs:
The Dean of Academic Affairs

Associate Prof. Dipl.-Ing.in Dr.in techn. Hilda Tellioglu



Diploma Supplement

This Diploma Supplement follows the model developed by the European Commission, Council of Europe and UNESCO / CEPES. The purpose of the supplement is to provide sufficient independent data to improve the international transparency and fair academic and professional recognition of qualifications (diplomas, degrees, certificates etc.). In addition to the certificates of the named person this supplement provides a description of the nature, level, context, content and status of the studies.

1	Information identifying the holder of the qualification	
1.1	Family name(s)	Kenbeek
1.2	Given name(s)	Vincent Theo Willem
1.3	Date of birth (DD.MM.YYYY)	11.07.1996
1.4	Student identification number	12229949
2	Information identifying the qualification	
2.1	Name of qualification, title conferred *)	Diplom-Ingenieur (equivalent to Master of Science)
2.2	Main field(s) of study for the qualification	Computer Engineering
2.3	Name and status of awarding institution *)	Technische Universität Wien, österreichische staatliche Universität (TU Wien, Austrian public University)
2.4	Name and status of institution administering studies *)	Technische Universität Wien, österreichische staatliche Universität; Fakultät für Informatik (TU Wien, Austrian public University; Faculty of Informatics)
2.5	Language(s) of instruction / examination	German, English

3	Information on the level of the qualification					
3.1	Level of qualification	ISCED Code 7				
3.2	Nominal length of program	4 semester				
3.3	Access requirement(s)	Completion of bachelor program or equivalent				
4	Information on the contents and results gained					
4.1	Mode of study	Full-time study				
4.2	Program requirements	Compulsory modules (60 ECTS Credits) Elective modules (21 ECTS Credits) Free elective courses and Transferable Skills (9 ECTS Credits) Diploma thesis and defense (30 ECTS Credits) See http://tiss.tuwien.ac.at				
4.3	Program details and individual grades	See "Transcript of Records"				
Course	Subject of Examination	Type	Semester hours	ECTS Credits	Date (dd.mm.yyyy)	Grade
182.755	Advanced Digital Design	VU	3.0	4.5	31.01.2023	sufficient
182.754	Advanced Digital Design	LU	1.5	1.5	25.01.2023	good
182.756	Advanced FPGA Design	VU	3.0	3.0	30.01.2024	excellent
184.726	Advanced Multiprocessor Programming	VU	3.0	4.5	06.02.2025	satisfactory
360.251	Advanced Programming with C++	VU	2.0	3.0	17.01.2023	excellent
311.062	Advanced Topics in Philosophy, History, and Trends of Technology	VO	1.0	1.5	16.01.2024	excellent
181.145	Computer Aided Verification	VU	2.0	3.0	07.05.2025	good
183.585	Computer Vision	VU	3.0	4.5	14.05.2025	satisfactory
181.144	Computer-Aided Verification	UE	2.0	3.0	23.08.2024	satisfactory
	Diploma Thesis			27.0	05.11.2025	excellent
104.272	Discrete Mathematics	UE	2.5	5.0	01.02.2023	sufficient
104.271	Discrete Mathematics	VO	4.0	4.0	28.02.2025	sufficient
384.154	Embedded Systems in FPGAs	VU	3.0	4.5	02.07.2024	good
185.A93	Formal Methods in Computer Science	UE	2.0	3.0	02.02.2024	good
185.291	Formal Methods in Computer Science	VU	4.0	6.0	26.05.2025	satisfactory
194.154	Generative AI	VU	2.0	3.0	12.02.2024	satisfactory
182.696	Hardware Modeling	VO	1.5	1.5	30.03.2023	satisfactory
191.127	Industrial Hardware Verification	VU	2.0	3.0	24.07.2024	good

Course	Subject of Examination	Type	Semester hours	ECTS Credits	Date (dd.mm.yyyy)	Grade
182.753	Internet of Things	VU	4.0	6.0	17.02.2025	excellent
	Master examination			3.0	26.11.2025	excellent
183.660	Mobile Robotics	VU	3.0	4.5	26.06.2023	satisfactory
188.923	Model Engineering	VU	4.0	6.0	16.02.2024	good
194.076	Modeling and Simulation	VU	2.0	3.0	05.04.2023	excellent
194.057	Research Methods	VU	2.0	3.0	28.06.2023	good
384.182	SoC Architecture and Design	VU	2.0	3.0	03.09.2024	sufficient
182.763	Stochastic Foundations of Cyber-Physical Systems	VU	4.0	6.0	20.06.2024	excellent

4.4	Grading scheme and grade distribution guidance	<i>Austrian Grading:</i> "excellent" (1) "good" (2) "satisfactory" (3) "sufficient" (4) "insufficient" (5)	<i>Assessment:</i> Excellent performance (ECTS-Grade A) Generally good, some mistakes (ECTS-Grade B) Balanced, number of substantial mistakes (ECTS-Grade C) Performance corresponding to minimal criteria (ECTS-Grades D/E) Requirement of further work (ECTS-Grades FX/F)
4.5	Overall classification of the qualification *)	"bestanden" (pass)	
5	Information on the function of the qualification		
5.1	Access to further study	Doctoral program	
5.2	Professional status conferred	Diplom-Ingenieur, Access to academic professions according to professional regulations, diploma in the sense of the directive on the recognition of professional qualifications, 2005/36/EC.	
6	Additional information		
6.1	Additional information	not applicable	
6.2	Further information sources	http://www.tuwien.ac.at http://www.bmfwf.gv.at http://www.oead.ac.at	
7	Certification of the supplement		7.4 Official stamp 
7.1	Date	26.11.2025	
7.2	Signature / name  Elfriede Oberleitner		
7.3	Capacity	By order of the Vice Rector Academic Affairs	
* in original language (German)			

Post-secondary Education in Austria

- The Austrian post-secondary **university sector** (*Hochschulsektor*) consists of
 - Public universities (*Universitäten*), maintained by the state;
 - Private universities (*Privatuniversitäten*), operated by private organisations with state accreditation;
 - Maintainers of university of applied sciences degree programmes (*Fachhochschul-Studiengänge*) incorporated upon the basis of private or public law and subsidised by the state, with state accreditation (some of which are entitled to use the designation *Fachhochschule*);
 - University colleges of education (*Pädagogische Hochschulen*) maintained by the state or operated by private organisations with state accreditation;
 - The Institute of Science and Technology Austria;
 - Universities of philosophy and theology (*Philosophisch-Theologische Hochschulen*), operated by the Roman Catholic Church.
- The **non-university post-secondary sector** (*außeruniversitärer postsekundärer Sektor*) consists of
 - Midwifery colleges;
 - Medical-technical academies;
 - Military academies (*Militärische Akademien*);
 - Vienna School of International Studies (*Diplomatische Akademie*);
 - Certain training institutions for psychotherapists (*Psychotherapeutische Ausbildungseinrichtungen*);
 - Conservatories (*Konservatorien*).

The following text addresses exclusively the university sector.

Overall Structure of University Education

There are currently two different systems of degree programmes in Austria: an older system not based on the Bologna process and a newer one based on it.

- Under the auspices of the **older system** of diploma programmes (*Diplomstudien*), the first degree awarded is the diploma degree (*Diplomgrad*). An Austrian higher secondary school leaving certificate or its equivalent is the general qualification necessary for enrolling in a diploma programme; conclusion of a diploma programme entitles degree holders to enrol in doctoral programmes. A diploma degree (*Diplomgrad*) is awarded by Austrian universities after a course of study consisting of 240 to 360 ECTS credits. Full degree titles are gender specific designations: *Magister* for men; *Magistra* for women. Degree titles also include a general description of the field of study in which they were obtained, e.g. *Magister philosophiae*. In the fields of engineering, the degree titles are *Diplom-Ingenieur/in*. Degrees awarded in medicine and dentistry are exceptions to the above. The first degrees awarded after the completion of these degree programmes consisting of 360 ECTS credits are *Doctor medicinae universae* and *Doctor medicinae dentalis*, respectively.

Graduates of university of applied sciences programmes that consist of 240 to 300 ECTS credits are awarded, analogous to university studies, a university of applied science diploma degree (*Fachhochschul-Diplomgrad*) contingent upon discipline: either a *Diplom-Ingenieur/in (FH)* for fields of engineering or *Magister/Magistra (FH)* in other fields of study.

- The **new system** is based on the distinction between undergraduate and graduate studies. Upon completion of an undergraduate programme (*Bachelorstudium* at universities; *Fachhochschul-Bachelorstudiengang* at universities of applied sciences; *Studiengang* at university colleges of education; 180 ECTS credits), a bachelors degree (designation: "Bachelor of/in...") is awarded. Upon completion of a graduate programme (*Masterstudium* at universities comprising 120 ECTS credits or, respectively, *Fachhochschul-Masterstudiengang* at universities of applied sciences comprising 60 to 120 ECTS credits), a masters degree (designation: "Master of/in ...") is awarded. In the fields of engineering, the designation of the masters degree can also be "Diplom-Ingenieur/in".

Recipients of these diploma degrees from the old system or masters degree from the new system (including the ones awarded in both cases by the universities of applied sciences) are entitled to enrol in doctoral programmes (*Doktoratsstudium*) at universities. A doctoral degree with the designation either "*Doktor/in*" or "Doctor of Philosophy" (PhD) is awarded upon completion of a doctoral programme with a minimum duration of three years.

In addition to the degree programmes (*ordentliche Studien*) described above, non-consecutive certificate programmes (*außerordentliche Studien*) are offered, for example in the form of university programmes for further education (*Universitätslehrgänge*) or individual units / modules in scientific subjects, both at universities; certificate university of applied sciences programmes for further education (*Lehrgänge zur Weiterbildung*) at universities of applied sciences; and certificate university college programmes for further education (*Hochschullehrgänge*) at university colleges of education.

Diploma Programme (Diplomstudium = old system)

Admission to a diploma programme is granted upon the basis of the Austrian higher secondary school leaving certificate (*Reifezeugnis*), its foreign equivalent, or the successful completion of a special university entrance qualification examination (*Studienberechtigungsprüfung*). Students of compulsory lower schools who have completed additional schooling in the form of apprenticeships as skilled workers also may take a vocationally based examination acknowledged as equivalent to the higher secondary school leaving certificate (*Berufsreifeprüfung*). Admission to diploma programmes in the arts is based on aptitude ascertained by admission examinations. Admission to university of applied sciences diploma programmes may also take place upon the basis of previous vocational or technical experience and qualifications of applicants. In some fields of study (e.g. Human Medicine and Dentistry, and university of applied sciences diploma programmes) admission is based on a selective admission process.

A degree programme may be divided into stages (*Studienabschnitte*). The length of each stage of the degree programme as well as the areas of study (*Fächer*) and content required are articulated in curricula that distinguish between required subjects (*Pflichtfächer*) and electives (*Wahlfächer*). Each stage concludes with a diploma examination (*Diplomprüfung*). University of applied sciences diploma programmes and some diploma programmes at universities include an internship or practical training. The approval of a diploma thesis (*Diplomarbeit*) is a prerequisite for admission to the concluding diploma examination.

Bachelor Programme (Bachelorstudium)

Admission to a bachelor programme is granted upon the basis of the Austrian higher secondary school leaving certificate (*Reifezeugnis*), its foreign equivalent, or the successful completion of a special university entrance qualification examination (*Studienberechtigungsprüfung*). Students of compulsory lower schools who have completed additional schooling in the form of apprenticeships as skilled workers may take a vocationally based examination acknowledged as equivalent to the higher secondary school leaving certificate (*Berufsreifeprüfung*). Admission to bachelor programmes in the arts is based on aptitude ascertained by admission examinations. Admission to university of applied sciences bachelor programmes is also possible on the basis of previous vocational or technical experience and qualifications of the applicants. In some university bachelor programmes, in most university of applied sciences bachelor programmes and in study programmes at university colleges of education, admission is based on a selective admission process.

Areas/modules of study (*Fächer/Module*) are specified in curricula. As a rule, the completion of two substantial bachelor papers or projects (*Bachelorarbeiten*) are required for awarding the degree. University of applied sciences bachelor programmes and some bachelor programmes at universities include an internship or practical training. The programme can conclude with a bachelor examination (*Bachelorprüfung*).

Master Programme (Masterstudium)

Admission to a master programme is granted on the basis of the successful completion of an Austrian bachelor programme (*Bachelorstudium*), or a comparable post-secondary degree acknowledged as equivalent. In some master programmes admission is based on a selective admission process.

Areas/modules of study (*Fächer/Module*) are specified in curricula. A core requirement is the completion of a master thesis (*Masterarbeit*). The approval of the master thesis (*Masterarbeit*) is a prerequisite for admission to this examination. This degree programme concludes with a master examination (*Masterprüfung*).

At university colleges of education no master programmes are offered.

Doctoral Programme (Doktoratsstudium)

Admission to a doctoral programme at a university is granted on the basis of the successful completion of an Austrian diploma or master programme, or a comparable post-secondary degree acknowledged as equivalent.

Contents and requirements of study are specified in curricula with the focus on the doctoral thesis (*Dissertation*) as the result of independent research. This degree programme concludes with the approval of the dissertation and with a comprehensive doctoral examination (*Rigorosum*) or a *defensio*.

At universities of applied sciences and at university colleges of education no doctoral programmes are offered.

Evaluation of performance and grading system (Austrian grading scale)

According to the modalities for examinations outlined in the curricula, achievement may be assessed by oral and written exams or project related work. In principle oral examinations are open to the public.

Austrian Grading scheme	Definition
1	EXCELLENT
2	GOOD
3	SATISFACTORY
4	SUFFICIENT
sucessfully completed	Positive performance, where a strict differentiation is not adequate
5	FAIL
not completed	Negative performance, where a strict differentiation is not adequate

Grades for comprehensive examinations, i.e. examinations covering materials from various subjects:

positive:

mit Auszeichnung bestanden (pass with distinction)
mit Erfolg bestanden (pass with merit)
bestanden (pass)

negative:

nicht bestanden (fail)

Source:

Federal Ministry of Science and Research

Unit III/7

November 2011